

Computer System Architecture - CSEN 601

Module 1: *Fundamentals of Computer Architecture*
Lecture 01: *Introduction to Computer Architecture*

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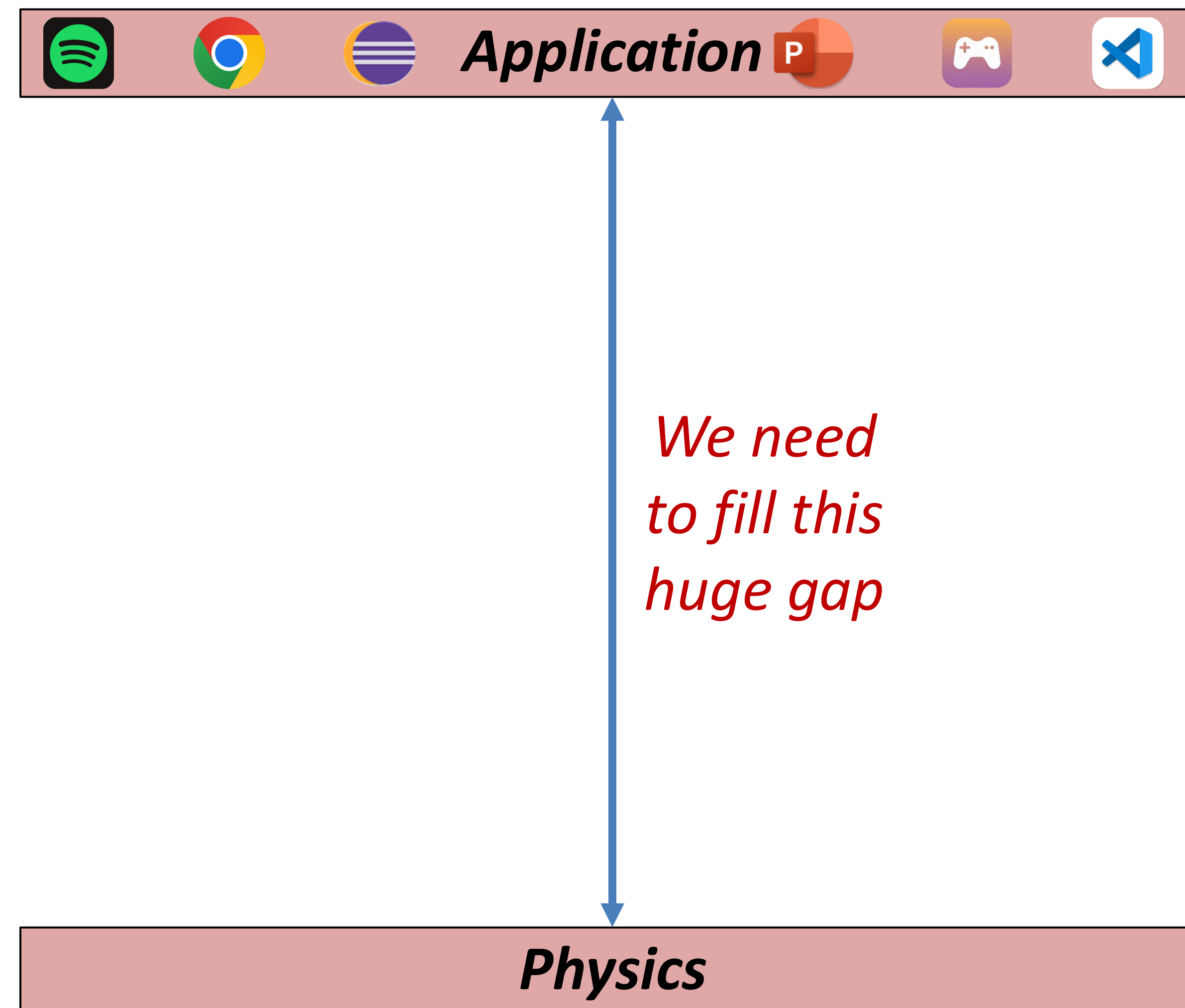
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- What is Computer Architecture?
- Abstractions in Modern Computing Systems
- Modern Computing Systems
- Computer Architecture Consideration
- The discipline of Computer Architecture
- The Formal Definition
- History of Computers
- Course Content

What is Computer Architecture?

- The Application Layer: the layer where something that a human wants to do lies, Example, calculate a spreadsheet, play a video game, play some sort of music, and we want to figure out how to map it down to physics.
- The Physics Layer: physics is what we have in the real world, Example, we have particles, and they bounce around, they interact, we have photons bouncing around in the world.

What we need is to translate what humans want to do into what the physics can do



What is Computer Architecture?

- That is what computer architecture is trying to do, to come up with

abstraction

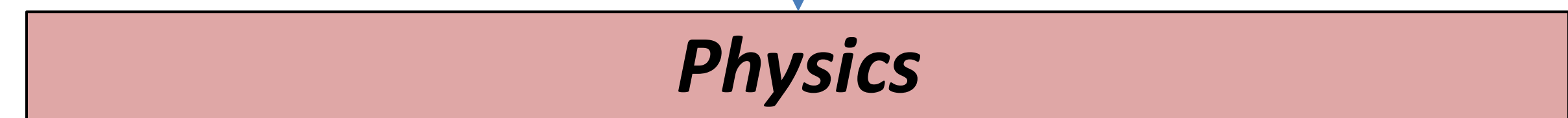
&

Implementation

layers to that allow us to execute information processing applications efficiently.

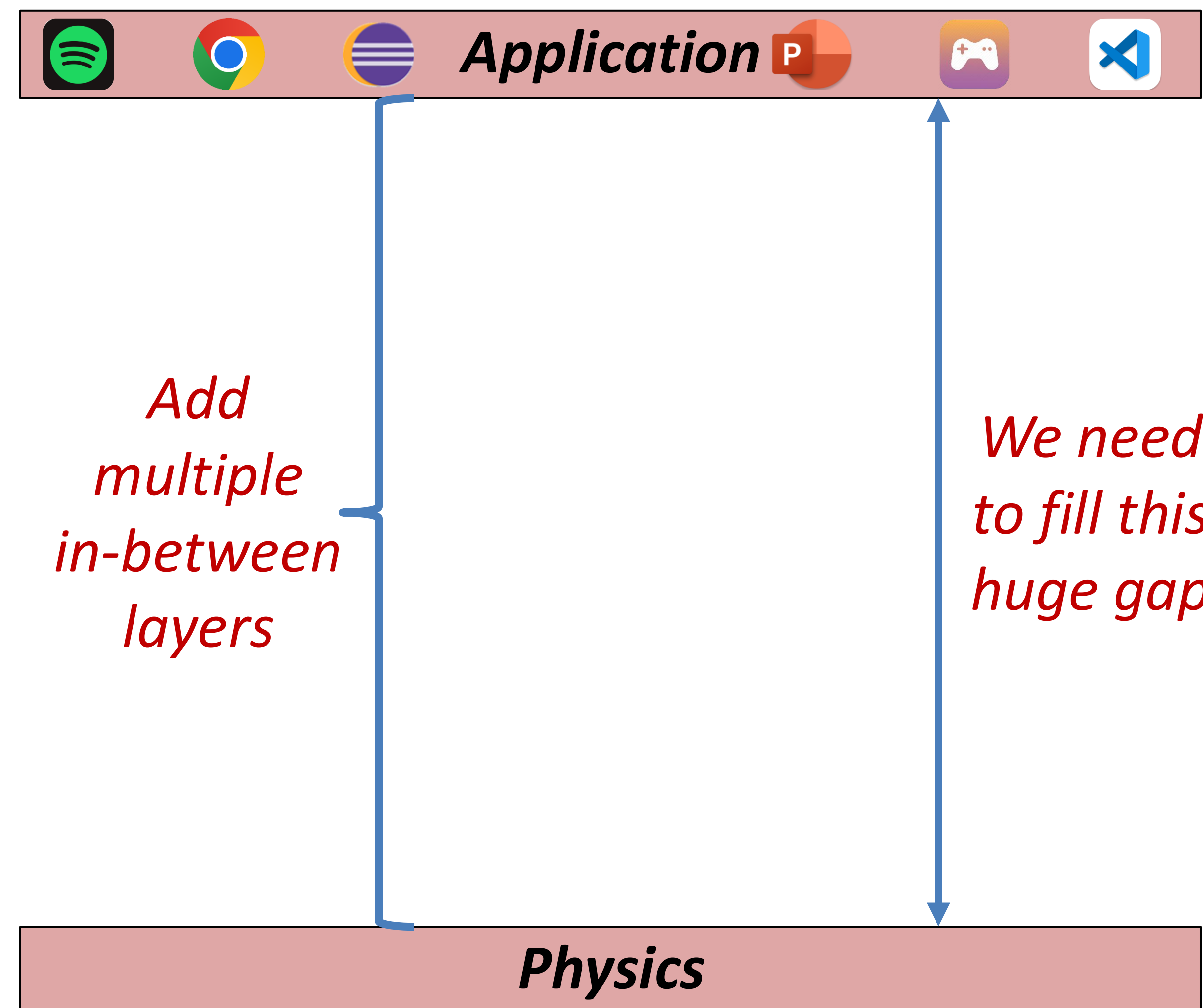


We need to fill this huge gap



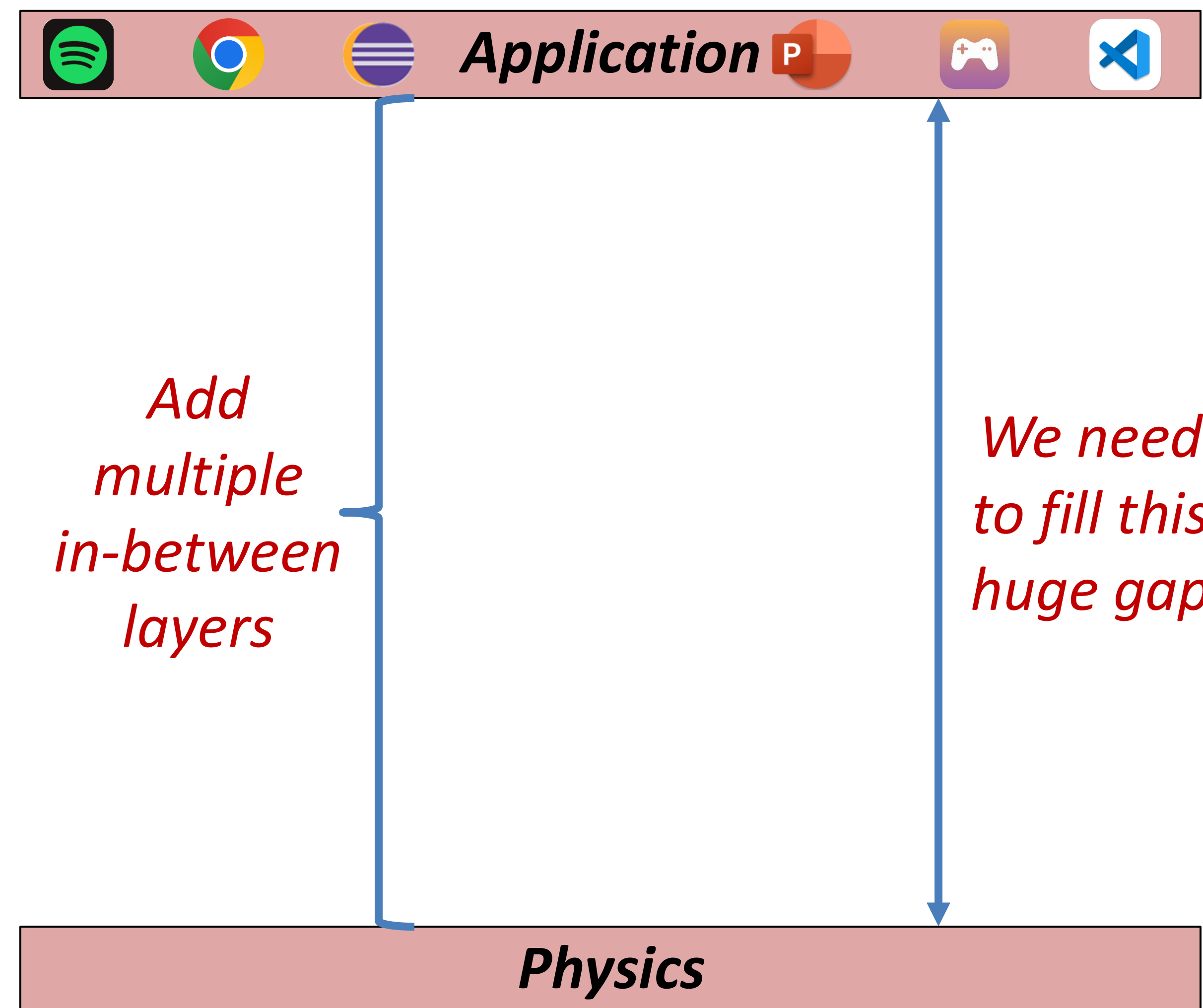
What is Computer Architecture?

- This can be done through adding several **in-between layers**, different in the functionalities.
- By adding these **smaller abstraction** layers and **implementation** layers, we can solve some of the gap's problems.
- Also, this will **ease the maintenance** in case of any problem in the intermediate layers, as now we do not have to change everything in the middle, just the one layer with the problem.



What is Computer Architecture?

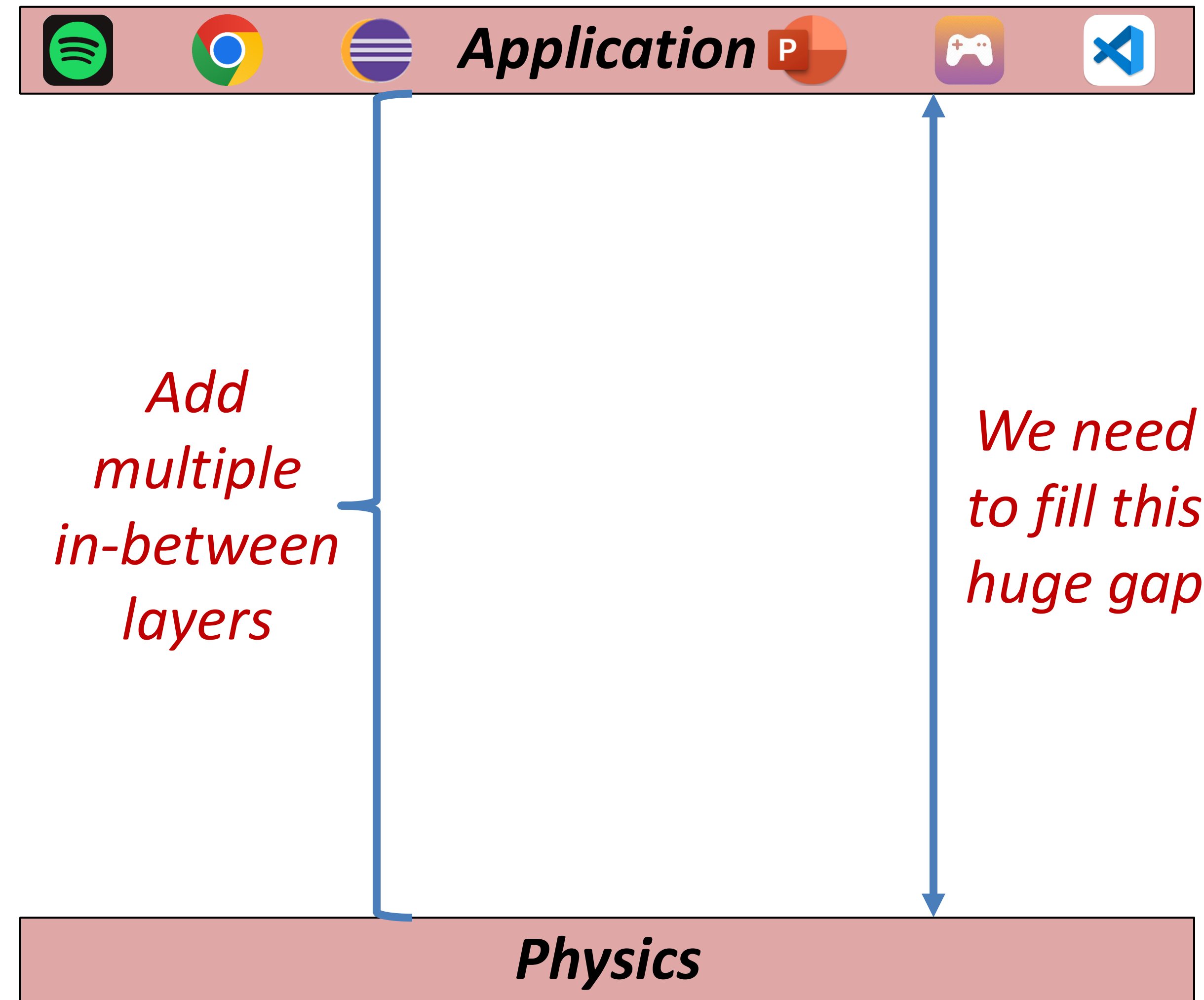
- Also, this will **help in the development** in case of any change in the application or the physics layers.
- Example: when physics changes, or we move to smaller transistors, or we move from, more advanced material as silicon to gallium arsenide or some other implementation technology, that changes the application layer, but we want to still be able to reuse a lot of the other work that we've done over the years.



What is Computer Architecture?

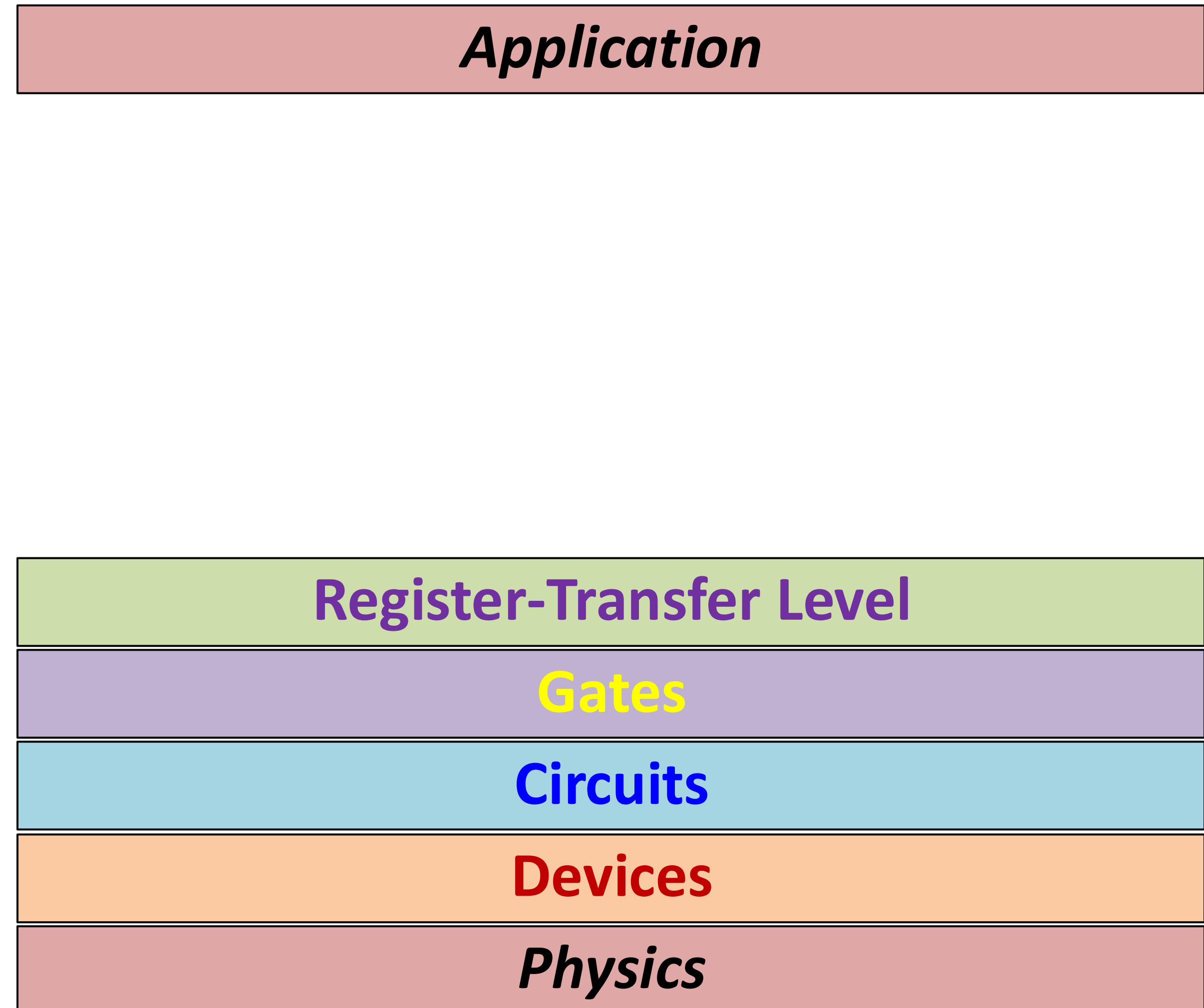
- Computer Architecture is the study of these layers, and figuring out how to take the physics and turn it into applications that humans want.
- This can be done through efficiently use manufacturing technologies.

We will be back later to this!!



Abstractions in Modern Computing Systems

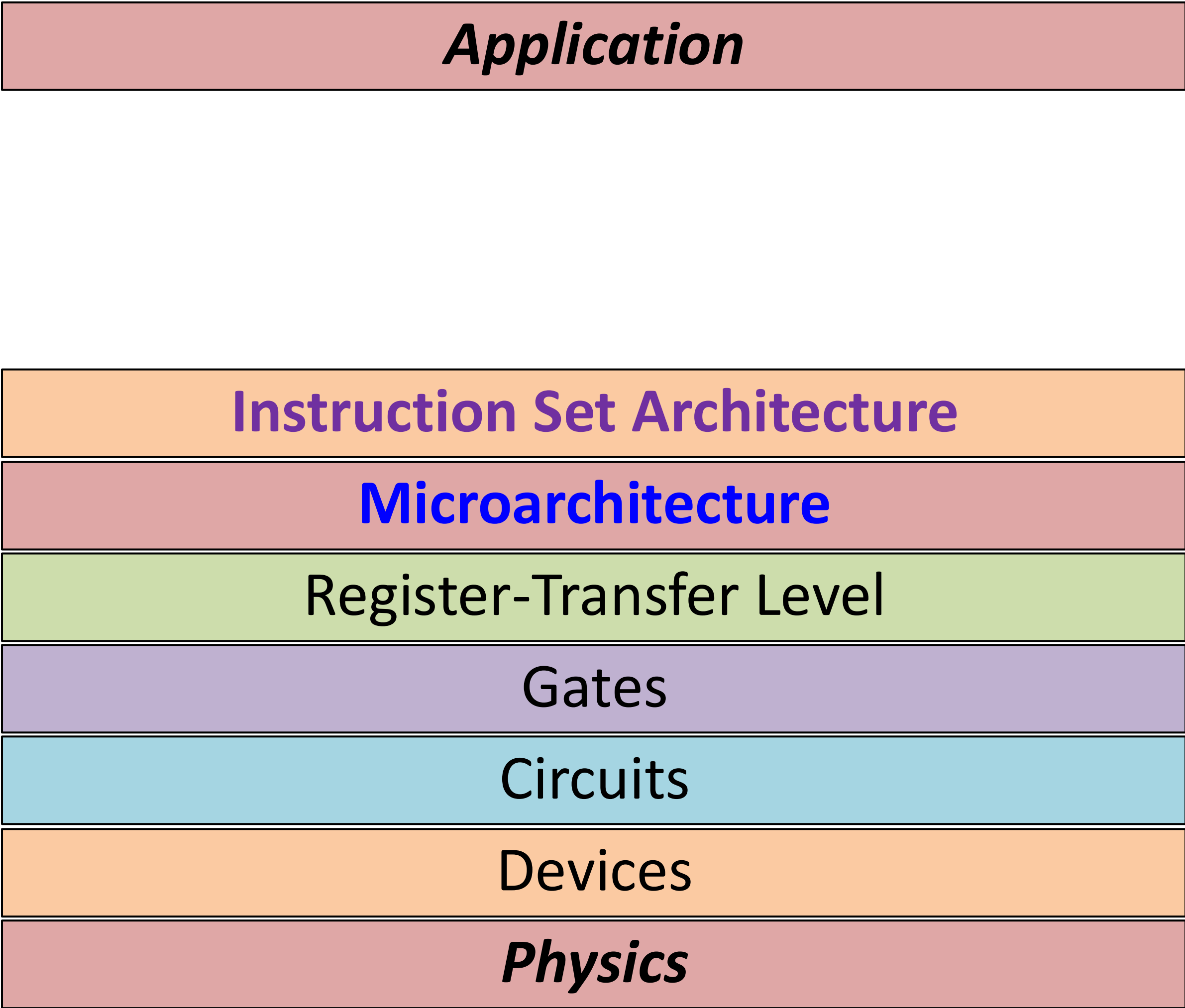
- Devices: They are the small electrical components such as **transistors** which can be used to build more advanced components such as Mosfets, Bipolar junction transistor (BJTs), Other types of FETs, and Others
- Using these devices, we start to build some bigger **Circuits**.
- Out of those we go and build **Gates**.
- And from gates, we can go to **Register-Transfer Levels (RTL)**, which is sort of our Verilog Coding; Hardware Description Language (HDL)



Abstractions in Modern Computing Systems

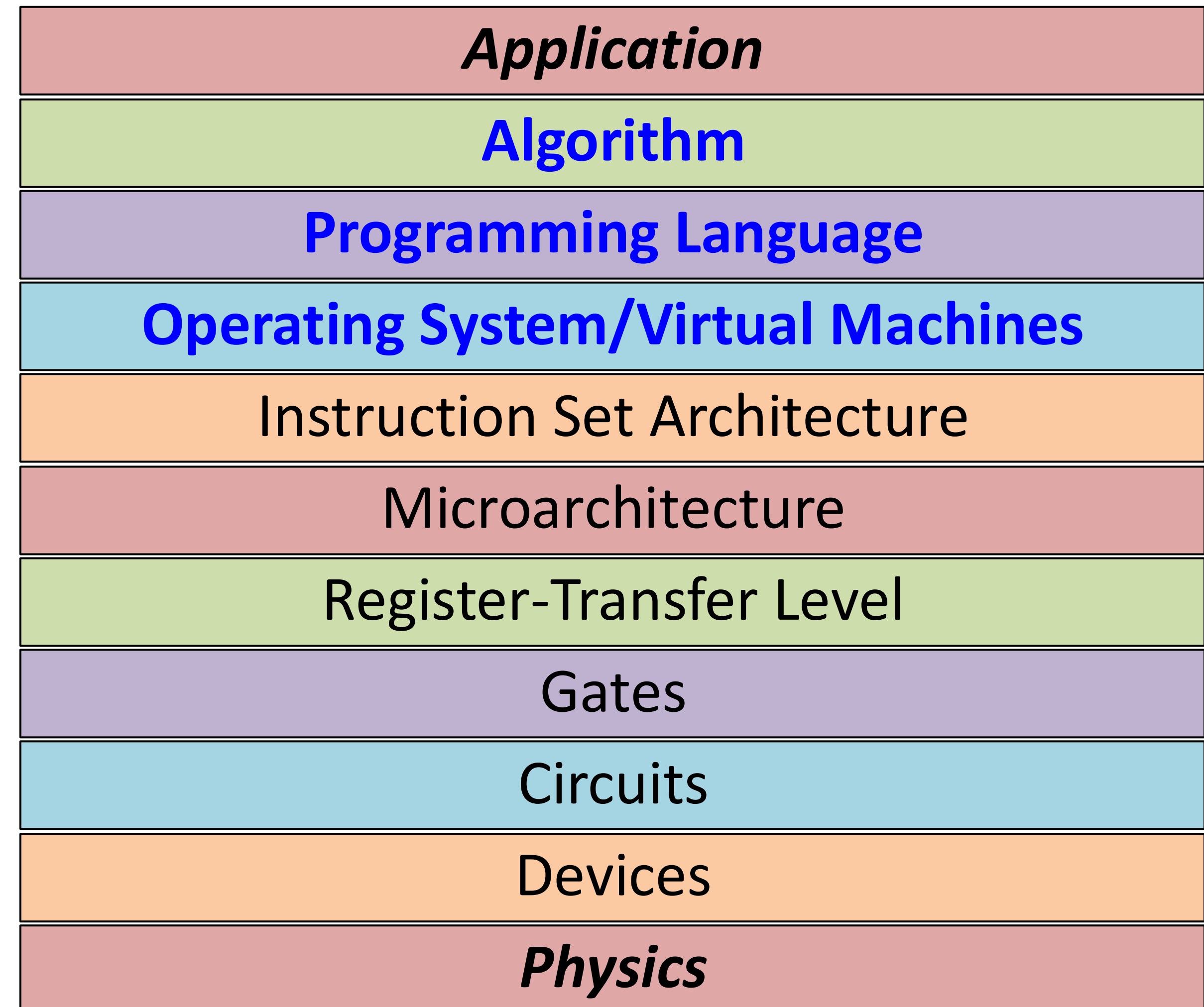
- Microarchitecture (μA): how you go and build a specific implementation of a chip.
- Instruction Set Architecture (big-A CA): gives some portability on top of the μA .

These are the core of our course



Abstractions in Modern Computing Systems

- Afterwards comes the Operating Systems and Virtual Machines which assists the abstraction between the H/W and Application layers,
- then Programming languages, and algorithms,
- and then finally the application programmer sitting on top.

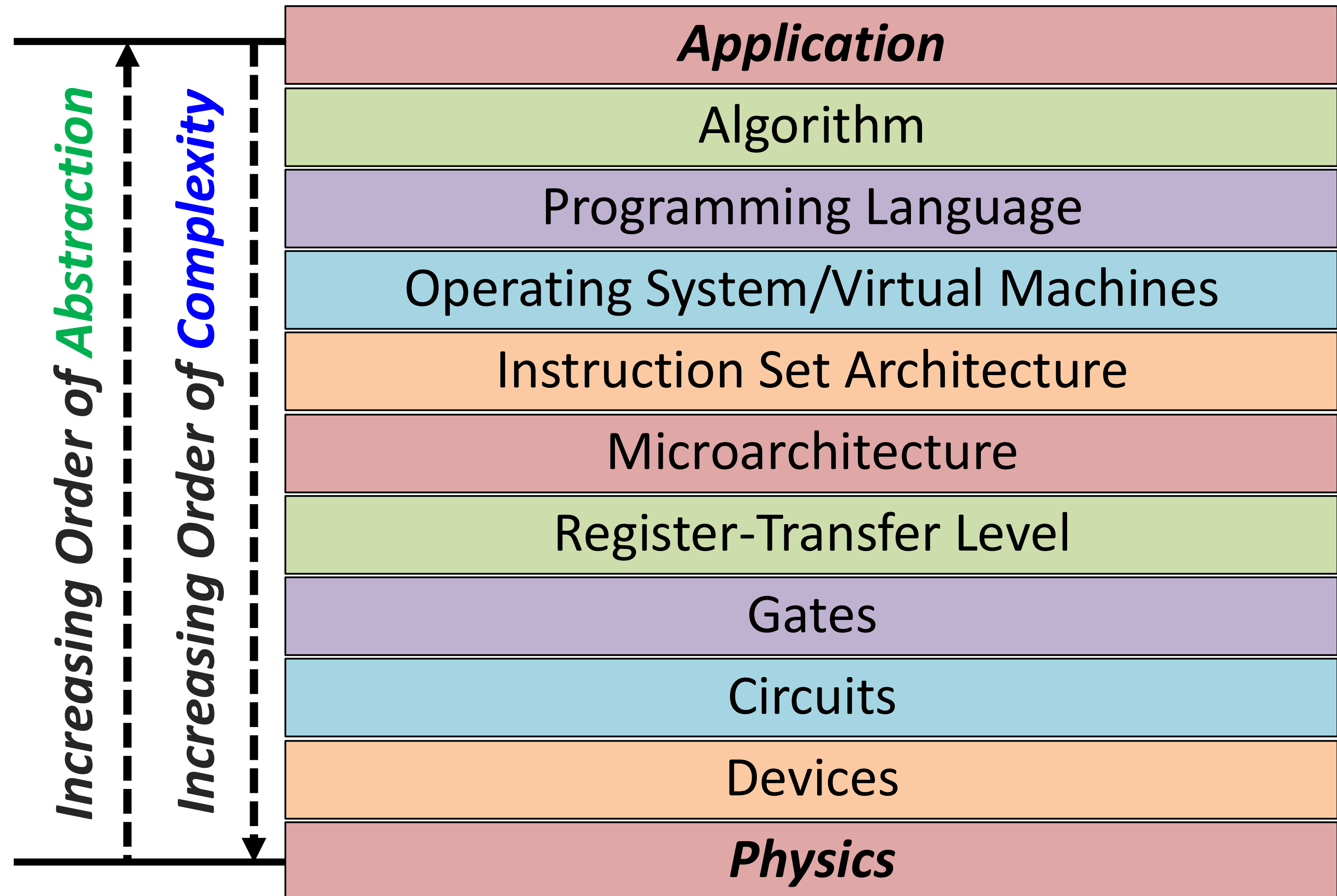


Modern Computing Systems

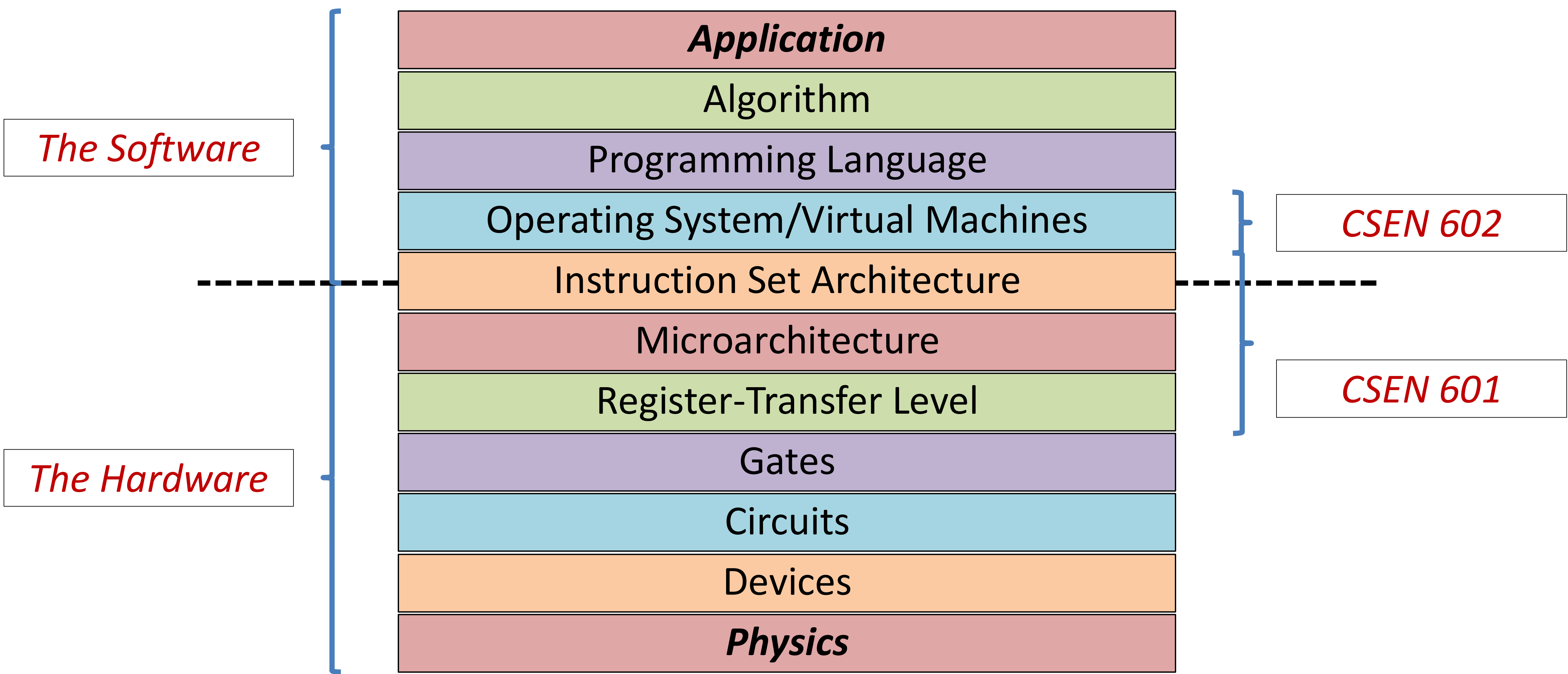
- It is worth to mention that when you traverse **upwards** the layers from the physics to the application layer, the **abstraction** level **increases**.

(remember the main role of the OS)

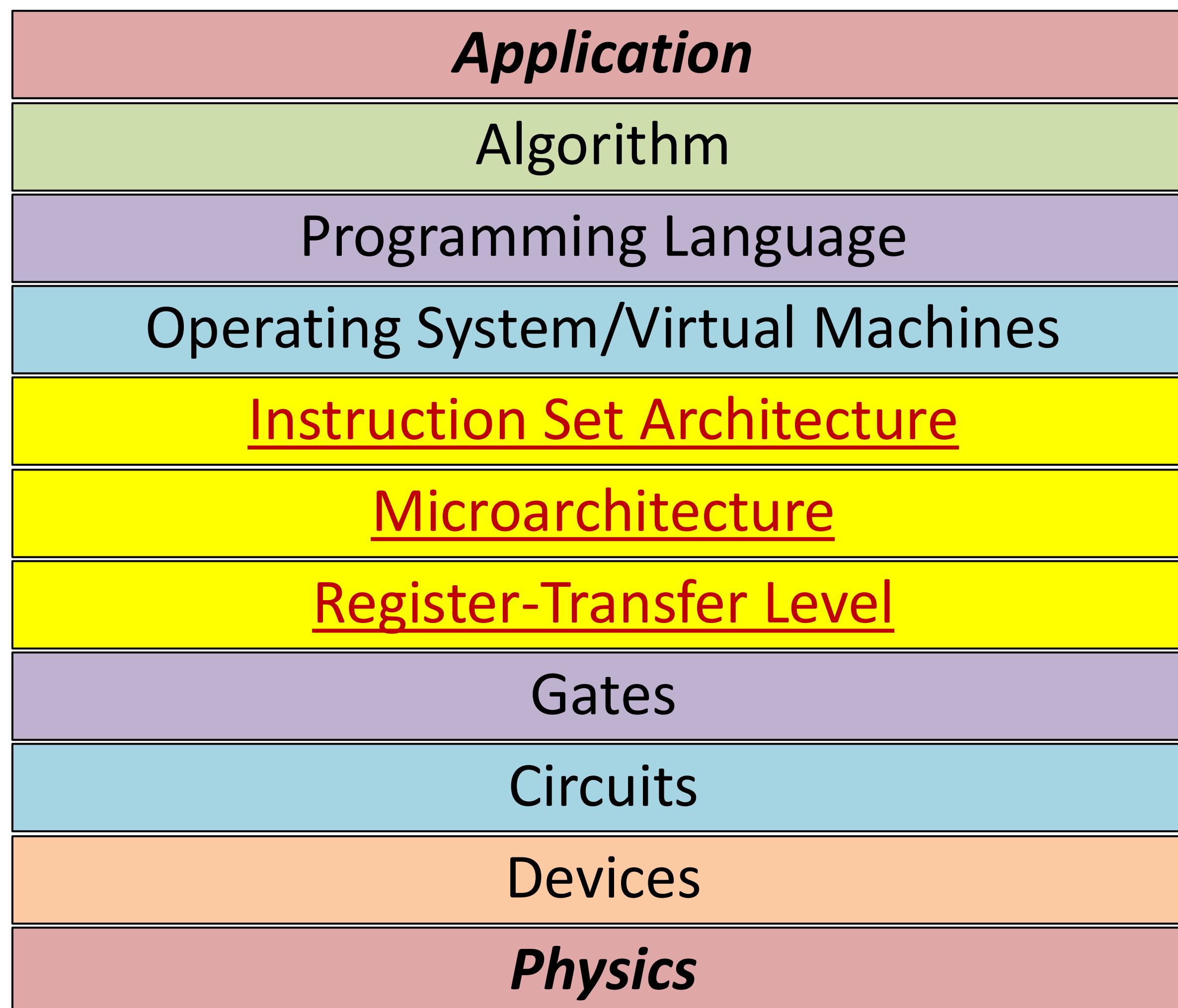
- Also, when you move **downwards** from application to the physics layer, the **complexity** of the layer **increases**.



Modern Computing Systems



Computer Architecture Consideration



Application Requirements:

- Suggest how to improve architecture
- Provide revenue to fund development

Architecture provides feedback to guide application and technology research directions

Technology Constraints:

- Restrict what can be done efficiently
- New technologies make new arch possible

The discipline of Computer Architecture

Instruction Set Architecture (ISA)

Defines the machine code that a processor reads and acts upon as well as the word size, memory address modes, processor registers, and data type.

Microarchitecture

Also known as "computer organization", this describes how a particular processor will implement the ISA. The size of a computer's CPU cache for instance, is an issue that generally has nothing to do with the ISA.

Systems Design/Register-Transfer Level

Includes all of the other hardware components within a computing system, such as data processing other than the CPU (e.g., direct memory access), virtualization, and multiprocessing.

In Conclusion

- Computer architecture refers to the design and organization of computer systems, encompassing both hardware and software components.
- It involves defining the structure and behavior of a computer system at various levels of abstraction, from the individual electronic components to the overall system organization.
- Computer architecture addresses how the different parts of a computer system work together to execute programs, process data, and perform computations efficiently and reliably.

The Generations



Vacuum Tube



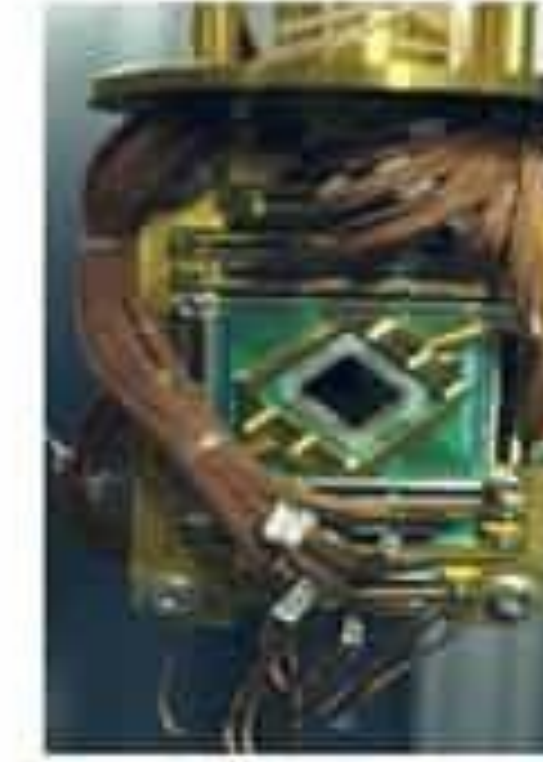
Transistors



Integrated Circuit



Microprocessor



**Quantum
Computer**



**1st Generation
Computer**



**2nd Generation
Computer**



**3rd Generation
Computer**



**4th Generation
Computer**

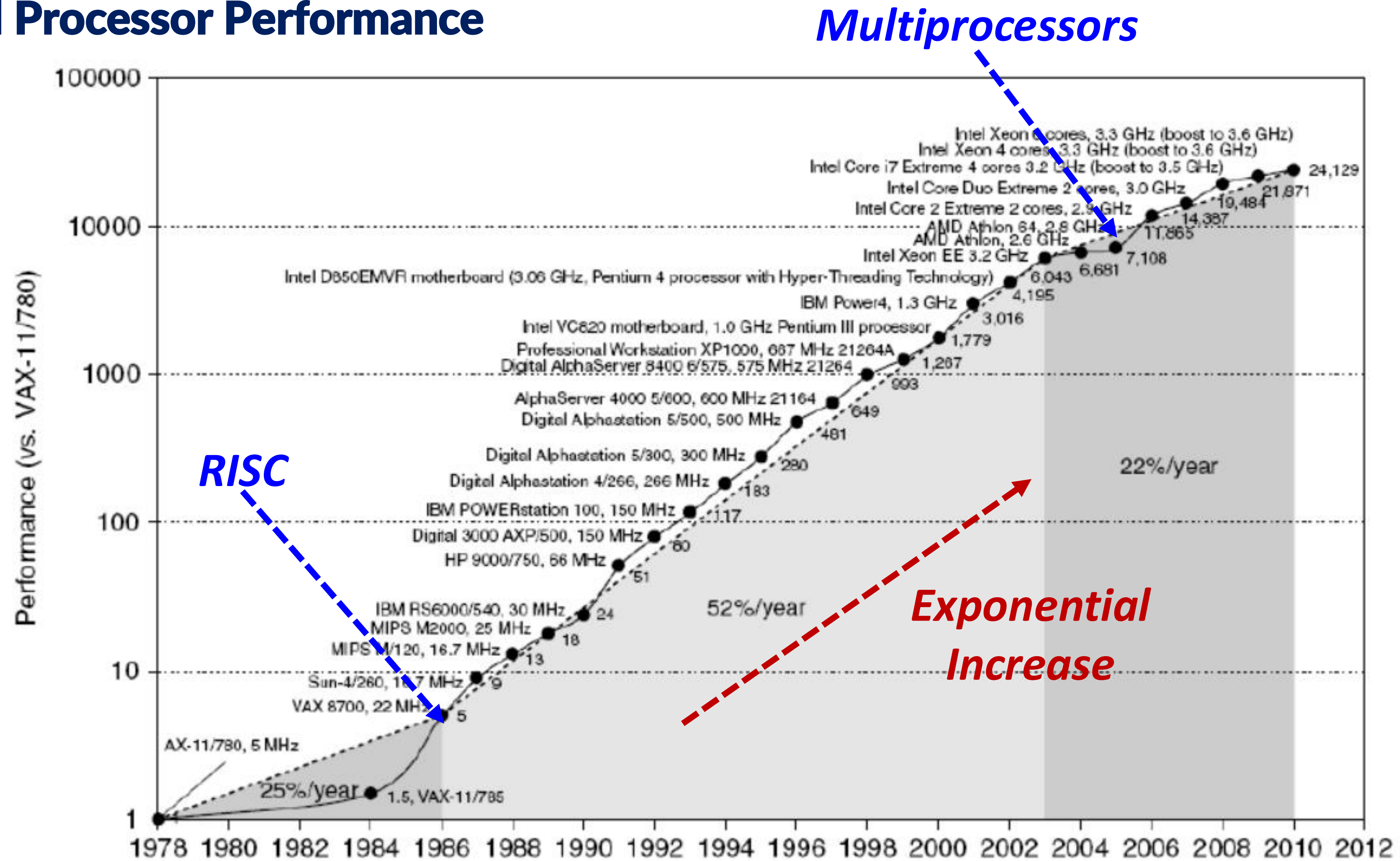


**5th Generation
Computer**

The Generations

	1 st Generation	2 nd Generation	3 rd Generation	4 th Generation	5 th Generation
Period	1940-1965	1956-1963	1964-1971	1971-Present	Present → Beyond
Circuitry	Vacuum Tube	Transistor	Integrated Circuit (IC)	Microprocessor (VLSI)	Ultra Large-Scale Integration
Memory Capacity	20KB	128KB	1MB	Semiconductor, high	ULSI & VLSI
Processing Speed	300 IPS	300 IPS	1MIPS	Faster than 3 rd Gen	Very Fast
Programming Language	Assembly	High-Level Language	C/C++	C/C++, Java	All Languages
Examples	UNIVAC, EDVAC	IBM 1401, CDC 3600	IBM 360 Series & 1900 Series	Pontium Series, Multimedia, stimulation	AI/Robotics

Sequential Processor Performance



4 Main Modules

- Module 1: Fundamentals of Computer Architecture
- Module 2: The Microarchitecture
- Module 3: The Instruction Set Architecture
- Module 4: Pipelining and Execution Techniques

For Further Inquiries, Please
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Thank you for your attention!

See you next time 😊